1. **Richardson extrapolation at infinity (I).**

   Consider the sequence

$$a_n = n \log\left(1 + \frac{1}{n}\right), \qquad n \geq 1.$$

   We wish to approximate

$$L = \lim_{n\to\infty} a_n.$$

   (a) Show that $L = 1$.

   (b) Show that, as $n \to \infty$,

$$a_n = L + \frac{c_1}{n} + \frac{c_2}{n^2} + \frac{c_3}{n^3} + O(n^{-4}),$$

   and determine c_1, c_2, and c_3.

   (c) Consider the extrapolated sequence

$$b_n = \alpha a_n + \beta a_{2n}.$$

   Determine α and β such that

$$b_n = L + O(n^{-2}).$$

   (d) Using b_n and b_{2n}, construct a sequence d_n satisfying

$$d_n = L + O(n^{-3}).$$

   Express d_n directly in terms of a_n, a_{2n}, and a_{4n}.

   (e) For $n = 10$, compute a_n, b_n, and d_n. Compare their absolute errors.

2. **Richardson extrapolation at infinity (II).**

   Consider the sequence

$$a_n = \sum_{k=1}^{n} \frac{1}{k^2}, \qquad n \geq 1.$$

   It is known that

$$L = \lim_{n\to\infty} a_n = \frac{\pi^2}{6}.$$

   Using the Euler–Maclaurin formula, one can show that

$$a_n = L - \frac{1}{n} + \frac{1}{2n^2} - \frac{1}{6n^3} + O(n^{-5}).$$

   (a) Construct a sequence b_n of the form

$$b_n = \alpha a_n + \beta a_{2n}$$

   such that

$$b_n = L + O(n^{-2}).$$

   Determine α and β.

(b) Using a_n, a_{2n}, and a_{4n}, construct a sequence d_n satisfying

$$d_n = L + O(n^{-3}).$$

(c) Can Richardson extrapolation be used to construct a sequence e_n satisfying

$$e_n = L + O(n^{-5})?$$

If so, express e_n as a linear combination of a_n, a_{2n}, a_{4n}, and a_{8n}.

(d) For $n = 10$, compute a_n, b_n, d_n, and e_n. Compare their absolute errors.

3. **Numerical differentiation by polynomial interpolation.**

Let $f \in C^3([x-h, x+h])$, where $h > 0$. Let p be the polynomial of degree at most two interpolating f at $x-h$, x, and $x+h$.

(a) Write p using the Lagrange interpolation formula and show that

$$p'(x) = \frac{f(x+h) - f(x-h)}{2h}.$$

(b) Recall that the polynomial interpolation error theorem states that, for every $t \in (x-h, x+h)$ distinct from the interpolation nodes, there exists $c_t \in (x-h, x+h)$ such that

$$f(t) - p(t) = \frac{f^{(3)}(c_t)}{6}(t-x+h)(t-x)(t-x-h).$$

Explain why the theorem does not guarantee that c_t is a differentiable function of t.

(c) For t distinct from the interpolation nodes, define

$$R(t) = \frac{6(f(t) - p(t))}{(t-x+h)(t-x)(t-x-h)}.$$

Show that R has a continuous extension to $t = x$, and determine $R(x)$ in terms of $f'(x)$ and $p'(x)$.

(d) Show that R is differentiable at $t = x$ and determine $R'(x)$ in terms of $f''(x)$ and $p''(x)$.

(e) Using the interpolation error theorem and the continuity of $f^{(3)}$, show that there exists $c \in [x-h, x+h]$ such that

$$f'(x) - p'(x) = -\frac{h^2}{6} f^{(3)}(c).$$

4. **Boole's rule, a five-point closed Newton–Cotes quadrature formula.**

(a) Verify that the following quadrature formula is exact for all polynomials of degree at most four:

$$\int_0^1 f(x)\,dx \approx \frac{1}{90}\left[7f(0) + 32f\left(\frac{1}{4}\right) + 12f\left(\frac{1}{2}\right) + 32f\left(\frac{3}{4}\right) + 7f(1)\right].$$

(b) Using the formula in part (a), derive a quadrature formula for

$$\int_a^b f(x)\,dx$$

that is exact for all polynomials of degree at most four.

(c) Use the formula from part (b) to approximate

$$I = \int_0^1 \frac{1}{t+1}\,dt.$$

Compare your approximation with the exact value of the integral and compute the absolute error.